

**TRACKING SUSTAINABILITY:
AN INSIGHT INTO WATER, CARBON AND
WEATHER FOOTPRINTS FOR A GREENER FUTURE**

A MAIN PROJECT REPORT

Submitted By:

732922CSR017

DEEPIKA N R

732922CSR031

ELANNGO M

732922CSR051

JANARTHANAN M

in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

VELALAR COLLEGE OF ENGINEERING AND TECHNOLOGY

ERODE - 638012

ANNA UNIVERSITY: 600 025

MAY 2026

SPECIMEN

**TRACKING SUSTAINABILITY:
AN INSIGHT INTO WATER, CARBON AND
WEATHER FOOTPRINTS FOR A GREENER FUTURE**

A MAIN PROJECT REPORT

Submitted By:

732922CSR017

DEEPIKA N R

732922CSR031

ELANNGO M

732922CSR051

JANARTHANAN M

in partial fulfilment for the award of the degree of

**BACHELOR OF ENGINEERING
in
COMPUTER SCIENCE AND ENGINEERING
VELALAR COLLEGE OF ENGINEERING AND TECHNOLOGY
ERODE - 638012
ANNA UNIVERSITY: 600 025**

MAY 2026

BONOFIDE CERTIFICATE

Certified that this project report **TRACKING SUSTAINABILITY: AN INSIGHT INTO WATER, CARBON AND WEATHER FOOTPRINTS FOR A GREENER FUTURE** is the bonafide work of **DEEPIKA N R (22CSR017), ELANNGO M (22CSR031), JANARTHANAN M (22CSR051)** who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported here does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Mrs.B.MOHANAPRIYA
SUPERVISOR
Associate Professor,
Department of CSE,
Velalar College of Engineering
and Technology,
Thindal,Erode - 638012.

SIGNATURE

Dr. S.JABEENBEGUM,M.E.,Ph.D.,
HEAD OF THE DEPARTMENT
Professor,
Department of CSE,
Velalar College of Engineering
and Technology,
Thindal,Erode - 638012

Submitted for Semester Mini-Project viva-voce examination held on _____.

INTERNAL EXAMINER

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We are greatly indebted to many people who have contributed to our progress in the main project and have encouraged us right from the start to make our ideas a reality.

We express our heartfelt thanks to our honorable Secretary and Correspondent, Thiru S.D. CHANDRASEKAR, B.A., who provided all the facilities to build our main project.

We express our deep sense of gratitude and sincere thanks to our Principal Dr. M. JAYARAMAN, M.E., Ph.D., for permitting us to undertake this main project.

We are indebted to our Dean, Prof. P. JAYACHANDAR, M.E., for giving us the opportunity and continuous inspiration to carry out the main project.

We express our profound gratitude to our beloved Head of the Department Prof. Dr. S. JABEEN BEGUM, M.E., Ph.D., for her valuable guidance and immensely profited by her comments and reviews.

We express our sincere thanks to our main Project Coordinator, Mrs. M. NANTHINI, B.E., M.E., Assistant Professor, for his guidance in doing our project successfully.

Our sincere thanks to our Supervisor, Mrs. B. MOHANAPRIYA B.E., M.E, for valuable guidance, advice, and help rendered whenever we approached in times of need.

We express our sincere thanks to our Parents for their support in doing our main project successfully.

ABSTRACT

Sustainability has become a critical global priority due to increasing environmental degradation, climate change, and resource depletion. The project titled **Tracking Sustainability: An Insight into Water, Carbon and Weather Footprints for a Greener Future** aims to develop an integrated framework for measuring and analyzing environmental performance using water footprint, carbon footprint, and weather-related indicators. The system combines environmental data analytics, sustainability metrics, and AI/ML-based modeling to provide a comprehensive assessment of ecological impact. By presenting results through interactive dashboards and decision-support tools, the project enables effective monitoring, prediction, and strategic planning for sustainable development.

Existing systems primarily focus on individual environmental indicators such as carbon emissions or water usage independently, without integrating multiple footprint dimensions into a unified framework. Most studies rely on historical or static datasets and provide descriptive analysis rather than predictive insights. Additionally, there is limited integration of climate variability and weather efficiency into sustainability evaluation models. These fragmented approaches restrict holistic decision-making and reduce the effectiveness of long-term environmental planning and policy implementation.

To overcome these limitations, the proposed system introduces an integrated sustainability assessment model that combines water, carbon, and weather footprints into a composite analytical framework. The system incorporates data preprocessing, footprint calculation, AI/ML-based predictive modeling, and visualization dashboards to generate actionable insights. It supports risk identification, trend forecasting, and sustainability performance ranking. The major advantages include holistic environmental evaluation, improved decision support, predictive sustainability planning, and adaptability for smart cities and sector-specific applications. This integrated and intelligent framework contributes toward building a greener, data-driven, and sustainable future.

TABLE OF CONTENTS

CHAPTER NO	TITLE	PG NO
	ABSTRACT	
	LIST OF TABLES	VII
	LIST OF FIGURES	VIII
	LIST OF ABBREVIATION	IX
	INTRODUCTION	
	1.1 INTRODUCTION	1
	1.2 PROBLEM STATEMENT	1
	1.3 FOOTPRINT	2
	1.3.1 WATER FOOTPRINT	2
1	1.3.2 CARBON FOOTPRINT	2
	1.3.3 WEATHER FOOTPRINT	2
	1.4 APPLICATIONS	4
	1.5 OBJECTIVE	5
	LITERATURE SURVEY	6
	2.1 Carbon Footprint of Conventional and Online Alternatives of National	6
2	Scientific Student Conference in Hungary	
	2.2 Examining the Carbon Footprint of Conferences with an	6

Emphasis on Energy Consumption and Catering	
2.3 Water Footprint as A Sustainability Indicator: A Systematic Literature Review	6
2.4 A Bibliometric and Systematic Review of Carbon Footprint Tracking in Cross-Sector Industries: Emerging Tools and Technologies	7
2.5 Evaluation of Personal Ecological Footprints for Climate Change Mitigation and Adaptation	7
2.6 Greening Our Future: A Comprehensive Analysis of Sustainable Practices and Their Role in Promoting Environmental Quality and Human Well-Being	8
2.7 Integrated Footprint Assessment and Holistic Sustainability Metrics	8
2.8 Research GAPS	9
2.9 EXISTING SYSTEM	9

METHODOLOGY / PROPOSED SOLUTION

3	3.1 OVERVIEW OF THIS PROJECT	10
	3.2 PROPOSED SYSTEM	10
	3.3 ADVANTAGES	11
	3.4 FUTURE LOOK	11
	3.5 SYSTEM ARCHITECTURE	12

	3.6 MODULE DESCRIPTION	15
4	RESULT	
	4.1 PREDICTED PERFORMANCE OUTPUT	17
	4.2 VISUAL OUTPUTS	18
5	CONCLUSION	
	5.1 CONCLUSION	19
	5.2 FUTURE ENHANCEMENT	19
	APPENDICES	20
	REFERENCES	42

LIST OF TABLES

TABLE NO	TABLE NAME	PAGE NO
2.8.1	RESEARCH GAPS	9

LIST OF FIGURES

FIGURE NO	FIGURE NAME	PAGE NO
3.5.1	SYSTEM ARCHITECTURE	13
3.6.7.1	DATA INTEGRATION	16
3.6.7.2	FOOTPRINT NEXUS	16
4.1.1	WEATHER FOOTPRINT	17
4.1.2	CARBON FOOTPRINT	17
4.1.3	WATER FOOTPRINT	18
4.2.1	AQUA TRACK FOOTRPRINT	18
4.2.2	ECO TRACK FOOTPRINT DASHBOARD	18

LIST OF ABBREVIATION

IoT	Internet Of Things
AI	Artificial Intelligence
ML	Machine Learning
CSR	Corporate Social Responsibility
GHG	Green Houe Gas
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analysis
CSI	Composite Sustainability Index

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Sustainability has become one of the most critical global priorities in the 21st century as the world faces increasing challenges related to climate change, environmental degradation, and depletion of natural resources. Human activities such as industrialization, urbanization, deforestation, and excessive consumption of water and energy have significantly disturbed the natural balance of the Earth. As a result, there is an urgent need to monitor and manage the impact of these activities on the environment in order to ensure a healthier and greener future for coming generations. The water footprint measures the amount of freshwater used in various processes such as agriculture, industries, and domestic consumption. With increasing population and industrial growth, water resources are becoming scarce, making efficient water management essential. The carbon footprint represents the total amount of greenhouse gases released into the atmosphere due to activities like transportation, electricity usage, and industrial production. Rising carbon emissions are the main cause of global warming and climate change. The weather footprint, which reflects changes in climatic conditions such as temperature, rainfall, and extreme weather events, highlights the long-term impact of environmental stress on the Earth's climate system. Tracking these three footprints together provides a comprehensive view of sustainability. By analyzing water usage, carbon emissions, and weather patterns, organizations, governments, and individuals can make informed decisions to reduce environmental harm and promote eco-friendly practices. The integration of digital technologies, data analytics, and environmental monitoring tools further enhances the ability to track and manage these footprints effectively.

This project, "Tracking Sustainability: An Insight into Water, Carbon, and Weather Footprints for a Greener Future," aims to study and analyze these key environmental indicators to promote awareness and encourage sustainable practices. By understanding the relationship between human activities and environmental impact, this work seeks to contribute toward building a cleaner, healthier, and more sustainable world.

1.2 PROBLEM STATEMENT

Rapid industrialization, urban growth, and increasing human activities have led to excessive consumption of natural resources and a significant rise in environmental pollution. Despite growing awareness about sustainability, most individuals and organizations lack a clear and integrated system to monitor their water usage, carbon emissions, and weather-related environmental impacts. These parameters are usually measured separately, making it difficult to understand their combined effect on the environment. The absence of a unified tracking and

analysis system results in poor decision-making, inefficient resource usage, and delayed response to climate change. Without accurate data and real-time insights, it becomes challenging to identify unsustainable practices and take corrective actions. Therefore, there is a need for an intelligent and integrated platform that can track, analyze, and visualize sustainability indicators to support informed environmental management.

1.3 FOOTPRINT

1.3.1 WATER FOOTPRINT

The **water footprint** is an important indicator of environmental sustainability that measures the total amount of freshwater used directly and indirectly by individuals, organizations, or products. It represents how much water is consumed during various activities such as drinking, cooking, farming, industrial production, and electricity generation. Since freshwater is a limited and valuable resource, understanding and managing the water footprint is essential for sustainable development.

Water footprints are generally classified into three main types:

1. Blue Water Footprint

This refers to the amount of surface and groundwater used for activities such as irrigation, household consumption, and industrial processes. It includes water taken from rivers, lakes, and underground sources.

2. Green Water Footprint

This represents the rainwater stored in soil and used by plants, especially in agriculture and forestry. It plays a crucial role in crop production and food security.

3. Grey Water Footprint

This indicates the amount of freshwater required to dilute pollutants to acceptable environmental standards. It reflects the level of water pollution caused by industrial, agricultural, and domestic activities.

Monitoring the water footprint helps in identifying areas where water is being wasted or overused. For example, agriculture consumes the largest share of global freshwater, followed by industries and households. Excessive water use leads to groundwater depletion, drying of rivers, and water scarcity in many regions.

In this project, water footprint data is collected and analyzed to evaluate how different activities impact freshwater resources. By tracking water usage and pollution levels, the system helps users understand their consumption patterns and take necessary steps to reduce water wastage. Efficient water management not only conserves resources but also supports ecosystem balance and long-term sustainability.

1.3.2 CARBON FOOTPRINT

The **carbon footprint** refers to the total amount of greenhouse gases, mainly carbon dioxide (CO₂), released into the atmosphere as a result of human activities. These emissions come from daily activities such as electricity usage, transportation, industrial production, and waste disposal. Carbon footprint is measured in units of **carbon dioxide equivalent (CO₂e)**, which represents the combined effect of all greenhouse gases on global warming.

The primary sources of carbon emissions include:

- Burning of fossil fuels for electricity and heat
- Vehicle emissions from cars, buses, and airplanes
- Industrial manufacturing processes
- Deforestation and land use changes
- Waste decomposition in landfills

An increasing carbon footprint leads to serious environmental problems such as global warming, climate change, melting of glaciers, rising sea levels, and extreme weather events. Reducing carbon emissions is therefore essential to protect the planet and maintain ecological balance.

In this project, the carbon footprint is calculated based on energy consumption, fuel usage, and transportation data. By analyzing these factors, the system provides an estimate of greenhouse gas emissions produced by individuals or organizations. This allows users to understand the environmental impact of their activities and identify areas where emissions can be reduced.

Tracking carbon footprint helps in promoting sustainable practices such as using renewable energy, reducing fuel consumption, adopting energy-efficient technologies, and minimizing waste. Through continuous monitoring and awareness, the system supports efforts to create a low-carbon and environmentally responsible future.

1.3.3 WEATHER FOOTPRINT

The **weather footprint** represents the impact of human activities on climatic and weather patterns over time. It reflects how changes in temperature, rainfall, humidity, air pressure, and the frequency of extreme weather events are influenced by environmental pollution, deforestation, urbanization, and greenhouse gas emissions. Unlike the water and carbon footprints, which measure direct resource usage and emissions, the weather footprint captures the broader environmental consequences of these actions on the Earth's climate system.

Human activities such as burning fossil fuels, industrial pollution, and large-scale land-use changes disturb the natural balance of the atmosphere. These disturbances lead to rising global temperatures, irregular rainfall, heat waves, floods, droughts, and cyclones. The weather footprint helps in understanding how these activities contribute to climate instability and long-term weather changes.

In this project, the weather footprint is analysed using meteorological data such as temperature trends, rainfall patterns, and frequency of extreme weather events. By comparing historical and current weather data, the system identifies climate variations and their relationship with human activities. This allows users to understand how environmental behaviour affects regional and global weather conditions. Monitoring the weather footprint supports better climate planning and disaster preparedness. It helps governments, organizations, and communities take preventive measures against climate risks and encourages the adoption of sustainable practices. By reducing water and carbon footprints, the weather footprint can also be minimized, contributing to a more stable and healthier climate for future generations.

1.4 APPLICATIONS

The sustainability tracking system has wide applications across different sectors where environmental monitoring and responsible resource management are essential. By analyzing water, carbon, and weather footprints, the system supports informed decision-making and promotes eco-friendly practices.

1. Smart Cities

The system can be used in smart cities to monitor water consumption, energy usage, and carbon emissions. City planners can identify areas of high resource usage and implement sustainable urban development strategies.

2. Industries and Manufacturing Units

Industries can track their water and carbon footprints to ensure compliance with environmental regulations. The system helps in optimizing production processes, reducing emissions, and improving environmental performance.

3. Agriculture and Irrigation Management

Farmers and agricultural organizations can use the system to monitor water usage and weather patterns. This enables efficient irrigation planning, crop selection, and climate-smart farming practices.

4. Educational Institutions

Schools and universities can use the platform to create awareness among students about sustainability. It can also be used as a learning tool for environmental science, data analysis, and climate studies.

5. Government and Environmental Agencies

Government bodies can use the system for policy formulation, environmental monitoring, and climate change analysis. It helps in assessing environmental impact and designing sustainable development programs.

6. Corporate Sustainability Management

Companies can track their environmental performance and generate sustainability reports. This supports corporate social responsibility (CSR) initiatives and improves public trust.

7. Disaster Management and Climate Planning

By analyzing weather footprint data, authorities can predict climate trends, prepare for extreme weather events, and reduce disaster risks.

1.5 OBJECTIVE

The main objective of this project is to design and develop an integrated system that tracks the water footprint, carbon footprint, and weather footprint in a unified platform. The project aims to collect, process, and analyse environmental data from multiple sources in order to provide accurate and meaningful insights into sustainability. By presenting this information through clear visualizations and detailed reports, the system helps users easily understand their environmental impact. In addition, the system is designed to monitor trends over time, identify areas of excessive water and energy usage, and provide useful recommendations for improvement. Through real-time tracking and historical analysis, the project supports environmental transparency, compliance with sustainability goals, and long-term sustainable development by identifying high-impact areas where corrective actions can lead to a greener and healthier future.

- To develop a user-friendly platform for real-time monitoring of sustainability indicators.
- To support environmental compliance and sustainability reporting.
- To encourage eco-friendly behaviour through data-driven insights.

CHAPTER 2

LITERATURE SURVEY

2.1 Carbon Footprint of Conventional and Online Alternatives of National Scientific Student Conference in Hungary

Carbon footprints, which address carbon dioxide or, simply, carbon, are a measure of the environmental impact of human activities. Carbon footprints can be connected to air pollution and climate change, and they indicate the total direct or indirect greenhouse gas (GHG) emissions caused by an individual, organisation, project, event, or product expressed in terms of carbon dioxide equivalent. The larger the carbon footprint of an activity, individual, community, or society, the higher the global warming impact (Tóthné Szita, 2018). The organization-level and event-level calculations of carbon footprints consider all the direct impacts and indirect components of events, which include operational emissions and vehicle emissions of a company; emissions generated by electricity, steam, and heat; transported and purchased products and services; carbon dioxide and greenhouse gas emissions of waste generation; and water supply.

2.2 Examining the Carbon Footprint of Conferences with an Emphasis on Energy Consumption and Catering

This research study presents a comparison of an in-person and an online conference in terms of environmental impact and energy efficiency. The main goal of our research was to prepare a complete life cycle assessment of a two-day (15-h), 200-participant in-person and online conference based on different impact assessment methods. Life cycle assessments focus on the numerical determination of the decarbonization of conference consumption (lunch, dinner, food and beverage consumption during breaks), conference organization (discussions, correspondence, abstract booklet, registration package), travel, and infrastructure. The meals were examined by connecting the stages of preparation, cooking, consumption and end-of-life cycle as a cradle-to-grave analysis. We paid particular attention to the calculation of energy consumption. After carbon footprint comparisons, the areas with the highest impacts with pie diagrams were identified. Lastly, a SWOT chart and an SAP-LAP analysis diagram summarize the achievable objectives and challenges. In conclusion, there is no outstanding difference between the impact assessment methods for the carbon footprint investigation.

2.3 Water Footprint as A Sustainability Indicator: A Systematic Literature Review

A comprehensive *bibliometric review* mapped global research trends in water footprint science, showing the shift from simple volumetric measures to *impact-oriented applications* tied to climate change and carbon emissions. Systematic analyses (semantic and thematic) show that water footprint research is increasingly linked to sustainability goals, revealing *hidden water use* in products and consumptive lifestyles and highlighting geographical and sectoral

differences. A specific *review on water stress and water footprint accounting* examines physical water stress induced by various water uses, noting the importance of integrating water scarcity metrics for sustainable management. Reviews from supply-chain perspectives highlight gaps in how water footprints are assessed across different industries and emphasize methodological challenges in pan-supply chain sustainability analyses.

2.4 A Bibliometric and Systematic Review of Carbon Footprint Tracking in Cross-Sector Industries: Emerging Tools and Technologies

In this study, we aim to investigate the technological advancements in carbon footprint tracking technologies in different industries and businesses, leveraging various approaches such as LCA, machine learning (ML), artificial intelligence (AI), Internet of Things (IoT), and blockchain. This review paper employs the bibliometric and scient metric analysis approach to identify the trending topics, keyword analysis, country analysis, and author analysis. The systematic review process follows a structured approach, which is divided into three main stages. This review study employs bibliometric analysis using statistical methods such as the Mann–Kendall test method and the Hurst exponent method. The methodology coheres with the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) framework and utilizes the SPICE (Setting, Perspective, Intervention, Comparison, Evaluation) method to develop research questions. It also makes use of bibliometric software such as VOSviewer and Biblioshiny. In the first stage, a protocol is developed, and then research questions are formulated to finalize the search strategy and PRISMA framework. Secondly, the data are fed to statistical models and bibliometric analysis software such as VOSviewer and Biblioshiny to yield social network analysis and thematic maps analysis. Finally, the findings are summarized, and potential future research directions are developed.

2.5 Evaluation of Personal Ecological Footprints for Climate Change Mitigation and Adaptation

Mitigating climate change involves reducing or preventing greenhouse gases (GHGs) to limit their adverse effects on the environment. Ecological footprints are helpful for identifying GHG emissions sources, such as fossil fuel combustion, deforestation, and industrial processes, so policymakers can target the biggest contributors to global warming. A more comprehensive understanding of these sources can lead to more effective strategies for reducing emissions and enhancing carbon sequestration. A study of the ecological footprint can help measure progress in this transition, in terms of carbon emissions reduction and growth in the adoption of renewable energy sources [2]).

A crucial role is played by ecological footprint analysis in the transportation sector as well. Fossil fuel-powered vehicles contribute substantially to GHG emissions [3]. Identifying opportunities to promote public transit, cycling, walking, and electric vehicles (EVs) can be achieved by analysing the ecological footprint of transportation. A major contribution to mitigating climate change can be made by encouraging the development of sustainable

transportation infrastructure and implementing policies that reduce the use of fossil fuel-powered vehicles.

2.6 Greening Our Future: A Comprehensive Analysis of Sustainable Practices and Their Role in Promoting Environmental Quality and Human Well-Being

In this paper we view the biosphere as a mesh of interconnected ecosystems that differ in spatial scales and operate at different speeds. The biosphere is a self-organising regenerative entity.² The Anthropocene can thus be read as being the era when the demand humanity makes of the goods and services produced by the biosphere in a period exceeds its ability to meet it on a sustainable basis. We call the (ecological) gap between demand and sustainable supply the *Impact Inequality*.³ One reason the gap has not been noted in the received macroeconomics of growth and development and the economics of poverty is that both subjects have neglected to include ecosystems in an essential way in their accounts. Gross domestic product (GDP) and its distribution reflect gross incomes, they do not account for the diminution of ecosystems when overstretched by human activities.

2.7 Integrated Footprint Assessment and Holistic Sustainability Metrics

Sustainability has become a central global priority as environmental degradation, climate change, water scarcity, and resource depletion increasingly threaten ecological balance and human well-being. Traditional approaches to environmental assessment have largely focused on isolated indicators such as carbon emissions, water consumption, or land use, often evaluated independently of social and economic dimensions. However, growing scientific evidence suggests that environmental systems are highly interconnected, and single-indicator assessments are insufficient to capture the complexity of sustainability challenges. Integrated footprint assessment has therefore emerged as a critical approach that combines water, carbon, ecological, and climate-related metrics into a holistic framework. Such comprehensive models not only evaluate environmental quality but also consider their implications for human health, economic stability, and social equity. By linking environmental indicators with well-being outcomes, integrated sustainability metrics provide a more accurate and actionable basis for policy-making, resource management, and long-term development planning. This shift toward multi-dimensional sustainability assessment represents an essential step in promoting resilient ecosystems and improving overall quality of life for present and future generations.

2.8 Research GAPS

Research Gap	Current Situation in Literature	Identified Gap
Lack of Integrated Assessment Frameworks	Most studies examine environmental quality (air, water, carbon emissions) and human well-being (health, income, social equity) separately.	Limited research develops holistic models integrating environmental indicators with social and psychological well-being metrics into a unified framework.
Limited Long-Term Impact Studies	Many sustainability studies are short-term or cross-sectional in nature.	Insufficient longitudinal research assessing the long-term (10–20 year) environmental and health impacts of sustainable practices such as green urban planning.
Technology-Driven Monitoring	Basic sustainability reporting tools used	Limited integration of AI, IoT, and real-time monitoring systems

TABLE 2.8.1: RESEARCH GAPS

2.9 EXISTING SYSTEM

- The calculation of water, carbon, and weather footprints is mainly done through manual methods.
- To provide static reports instead of real-time monitoring and forecasting.
- Use of limited AI or ML, resulting in less accuracy and poor adaptability to complex environmental patterns.
- Weak policy - performance linkage with footprint data.
- No unified sustainability index combining water, carbon, and climate risk.

CHAPTER 3

METHODOLOGY / PROPOSED SOLUTION

3.1 OVERVIEW OF THIS PROJECT

This project focuses on developing a comprehensive sustainability assessment framework by analyzing three critical environmental dimensions: water footprint, carbon footprint, and weather (climate) impact factors. Traditional sustainability monitoring systems evaluate these components separately, which limits a complete understanding of environmental performance and its impact on human well-being. This project aims to integrate these indicators into a unified analytical model to provide a holistic sustainability evaluation.

The project involves collecting and analysing environmental datasets related to water consumption, carbon emissions, and climate variables such as temperature and rainfall. Using statistical analysis, trend evaluation, and predictive modelling techniques, the system identifies patterns, risk factors, and sustainability performance levels. The study also explores how climate variability influences water and carbon footprints, helping to understand long-term environmental resilience.

By combining multi-dimensional environmental metrics, this project seeks to:

- Assess sustainability performance more comprehensively
- Identify high-impact sectors contributing to environmental degradation
- Analyse trade-offs between water use, carbon emissions, and climate variability
- Provide data-driven recommendations for policy-makers and stakeholders

The final outcome of this project is an integrated sustainability tracking framework that supports informed decision-making, promotes environmental quality, and contributes to a greener and more resilient future.

3.2 PROPOSED SYSTEM

The proposed system introduces an Integrated Sustainability Tracking Framework that combines Water Footprint (WF), Carbon Footprint (CF), and Weather/Climate Indicators into a unified analytical platform. Unlike existing systems that assess these indicators independently, the proposed model integrates multi-dimensional environmental data to provide a comprehensive sustainability performance evaluation.

The system is designed to enable:

- Real-time and historical sustainability tracking
- Multi-footprint comparison and trade-off analysis
- Climate-resilience evaluation
- Data-driven policy and strategic decision-making
- Explore AI/ML-driven platform with real-time data integration with High Accuracy.
- Provides Location-specific data for relevant results.
- The Footprint Trinity: Combining Weather, Carbon, and Water into a Single Perspective.

3.3 ADVANTAGES

Integrated Multi-Dimensional Assessment :

- Eliminates fragmented sustainability analysis.
- Provides a holistic environmental performance evaluation.

Predictive and Trend-Based Insights :

- Identifies long-term sustainability trends.
- Predicts future environmental risks.

Promotes Environmental Quality & Human Well-Being :

- Supports continuous sustainability tracking and public health-oriented sustainability planning.
- Encourages balanced ecological and social development.

Real-Time Monitoring Potential :

- Can integrate IoT and AI-based monitoring systems.
- Enhances transparency in reporting.

3.4 FUTURE LOOK

The future development of this project lies in transforming it into a comprehensive, intelligent, and globally adaptable sustainability monitoring ecosystem. The system can be upgraded to incorporate real-time data collection through IoT sensors, satellite data, and smart environmental monitoring devices, enabling continuous tracking of water usage, carbon emissions, and climate conditions instead of relying solely on historical datasets. By integrating advanced AI and machine learning models, the framework can predict future carbon and water footprint trends, forecast climate-related environmental risks, identify high-risk regions or sectors, and recommend optimized sustainability strategies, making the system proactive rather than reactive. Furthermore, the project can evolve into the creation of a standardized Composite Sustainability Index that integrates water footprint, carbon footprint, climate vulnerability indicators, and environmental health metrics, providing a unified decisionmaking tool for

governments, industries, and international organizations. The framework can also be extended to smart city infrastructure, urban sustainability dashboards, green building assessments, and sustainable transportation monitoring to support climate-resilient urban planning. Sector-specific applications in agriculture, industry, renewable energy, and IT/data centres can enhance practical implementation across multiple domains.

Additionally, incorporating policy simulation capabilities—such as carbon tax modelling, water regulation analysis, and climate adaptation planning—would enable policymakers to evaluate environmental strategies before implementation. Future enhancements may also integrate drought risk assessment, flood vulnerability modelling, heatwave impact analysis, and climate resilience scoring to strengthen adaptation planning. Finally, the system could contribute to global data standardization efforts, cross-country sustainability benchmarking, and integration with SDG monitoring platforms, positioning it as a globally collaborative sustainability intelligence framework.

3.5 SYSTEM ARCHITECTURE

Layer 1: Data Collection Layer

This is the input layer of the system.

Data Sources:

- Water usage datasets (agriculture, industry, urban)
- Carbon emission datasets (energy, transport, industry)
- Weather & climate data (temperature, rainfall, humidity)
- Environmental health indicators
- Public databases / APIs / Satellite data

This layer gathers structured and unstructured environmental data from multiple sources.

Layer 2: Data Preprocessing Layer

This layer prepares raw data for analysis.

Functions:

- Data cleaning (removing duplicates, handling missing values)
- Data normalization & scaling
- Outlier detection

- Feature selection
- Data transformation.

This ensures consistency and accuracy before footprint computation.

Layer 3: AI/ML Modeling Layer

This Layer is the **Intelligence Layer** of the system where Artificial Intelligence (AI) and Machine Learning (ML) techniques are applied to analyse environmental data, detect patterns, predict future trends, and support decision making. It transforms raw sustainability data into **predictive, optimized, and actionable insights**.

Functions:

- Predict future carbon and water footprint trends
- Forecast climate-related environmental risks
- Identify high-impact sectors or regions
- Detect anomalies in environmental data
- Recommend optimized sustainability strategies.

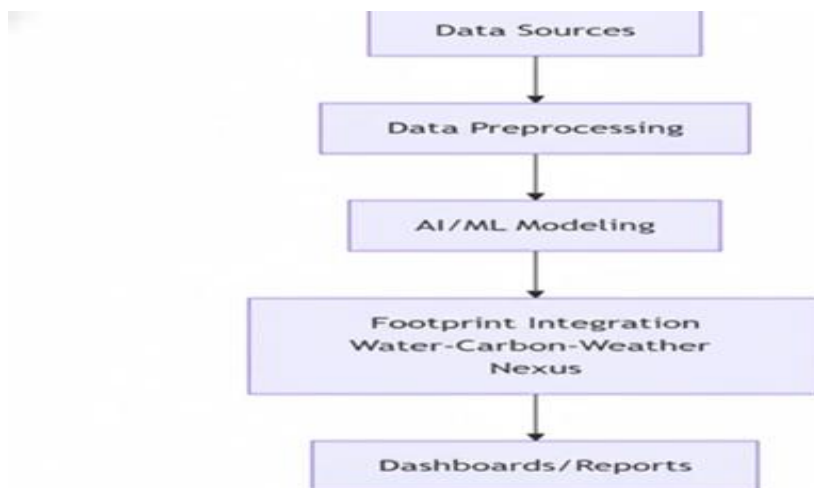


Fig 3.5.1 SYSTEM ARCHITECTURE

Footprint Calculation Engine

This is the core analytical layer.

Modules:

1. Carbon Footprint Calculator

- CO₂ equivalent calculation
- Emission factor application

2. Water Footprint Estimator

- Blue, Green, Grey water computation

3. Weather Impact Analyzer

- Climate variability correlation
- Risk factor modeling

This layer integrates environmental indicators into measurable sustainability metrics.

Layer 4: Integrated Sustainability Analysis Layer

This layer combines all footprint metrics.

Functions:

- Multi-dimensional footprint integration
- Trade-off analysis (carbon vs water vs climate)
- Sustainability performance scoring
- Composite Sustainability Index (CSI) generation

This produces a unified sustainability evaluation model.

Layer 5: Visualization & Decision Support Layer

This is the output layer.

Outputs:

- Interactive dashboards
- Sector comparison charts
- Sustainability index display
- Climate risk alerts
- Policy recommendation reports

Stakeholders can monitor sustainability performance in real-time.

3.6 MODULE DESCRIPTION

3.6.1. Data Collection Module

This module gathers environmental and sustainability-related data from reliable sources such as government databases, climate repositories, and open datasets. It collects information related to water consumption, carbon emissions, temperature, rainfall, and environmental indicators. The data may include both historical and real-time datasets (future scope with IoT integration). The collected data is stored in a structured format for analysis. This module forms the foundation of the entire sustainability assessment system.

3.6.2. Data Preprocessing Module

This module prepares the collected raw data for analysis and modeling. It handles missing values, removes duplicates, and corrects inconsistent entries. Data normalization and scaling are performed to maintain uniformity across variables. Outlier detection techniques are applied to improve data reliability. Proper preprocessing ensures accurate and meaningful analytical results.

3.6.3. Footprint Calculation Module

This module computes environmental impact metrics such as carbon footprint and water footprint. It uses standard formulas and sustainability indicators to quantify environmental performance. Climate-related variables are also analyzed to assess environmental stress levels. The calculated values help in measuring the sustainability status of regions or sectors. This module converts raw environmental data into measurable impact indicators.

3.6.4. AI/ML Modeling Module

This module applies machine learning algorithms to analyze sustainability patterns and trends. Regression models are used for predicting future carbon and water footprint levels. Classification techniques help identify high-risk and low-risk regions. Clustering methods group similar sustainability profiles together. This module enhances the system by making predictive and intelligent decisions.

3.6.5. Integrated Sustainability Index Module

This module combines multiple environmental indicators into a single composite sustainability score. It applies weighted scoring techniques to balance different footprint measures. The index reflects overall environmental quality and sustainability performance. It enables comparison

between regions, sectors, or time periods. This unified metric supports holistic sustainability evaluation.

3.6.6. Visualization & Dashboard Module

This module presents the analytical results in a clear and interactive format. It includes charts, trend graphs, heatmaps, and KPI indicators. Users can easily understand footprint levels and risk categories through visual tools. Filters and comparisons help analyze data across time and regions. This module supports effective communication of sustainability insights.

3.6.7. Decision Support & Recommendation Module

This module generates actionable sustainability recommendations based on analysis results. It identifies major contributing risk factors affecting environmental quality. Preventive strategies for carbon reduction and water conservation are suggested. It also supports policy-level planning and resource allocation decisions. This module transforms analytical outputs into practical environmental action plans.



FIG 3.6.7.1 (DATA INTEGRATION)



FIG 3.6.7.2 (FOOTPRINT NEXUS)

CHAPTER 4

RESULT

4.1 PREDICTED PERFORMANCE OUTPUT

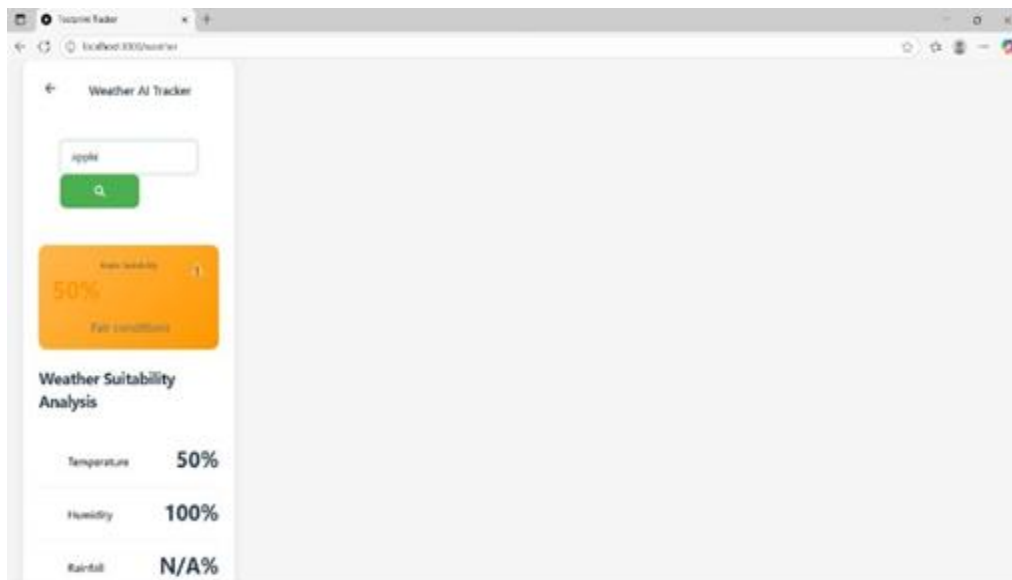


FIG 4.1.1 WEATHER FOOTPRINT

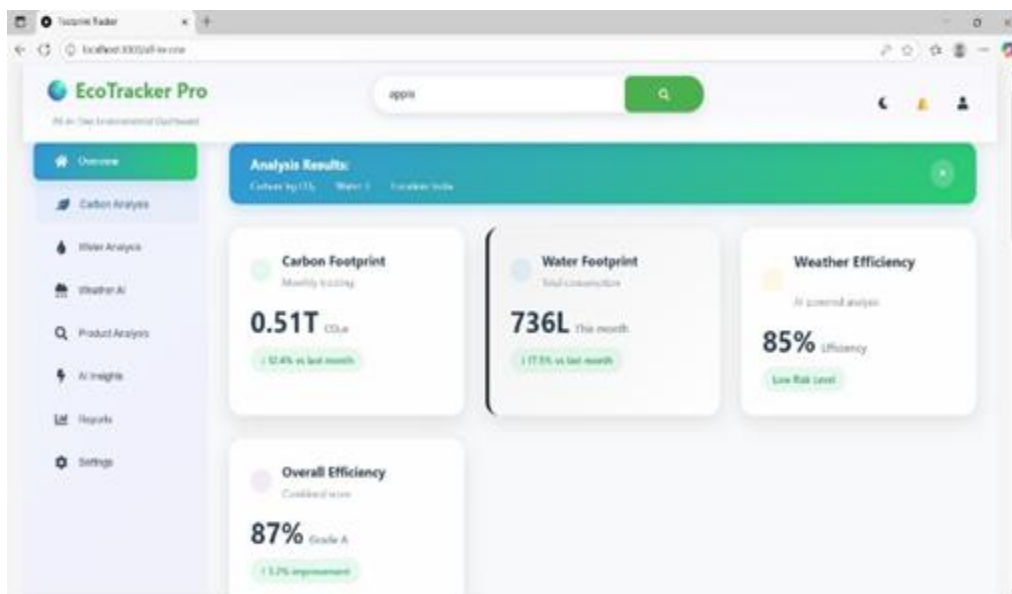


FIG 4.1.2 CARBON FOOTPRINT

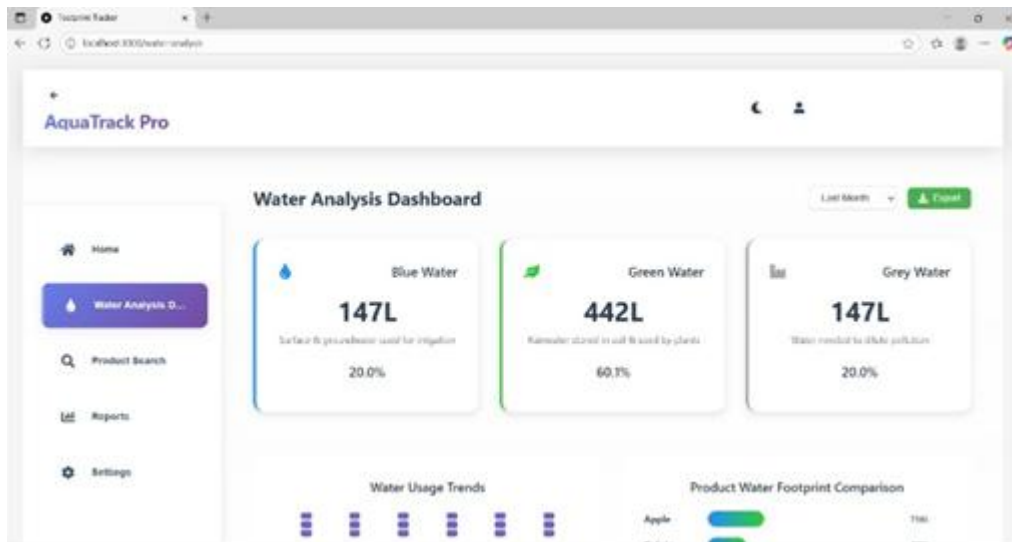


FIG 4.1.3 WATER FOOTPRINT

4.2 VISUAL OUTPUTS



FIG 4.2.1 AQUA TRACK FOOTPRINT

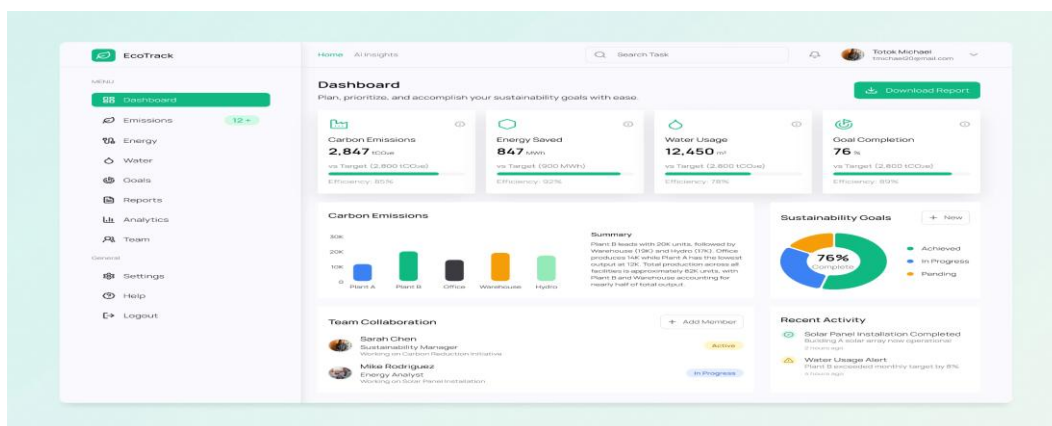


FIG 4.2.2 ECOTRACK FOOTPRINT DASHBOARD

CHAPTER 5

CONCLUSION

5.1 CONCLUSION

The project “Tracking Sustainability: An Insight into Water, Carbon and Weather Footprints for a Greener Future” presents a comprehensive framework for measuring and analyzing environmental sustainability using integrated footprint assessment. By combining water footprint, carbon emissions, and weather efficiency indicators, the system provides a holistic view of environmental performance rather than evaluating each factor separately. The incorporation of AI/ML techniques enhances predictive capabilities, enabling trend forecasting, risk identification, and intelligent sustainability insights.

The developed dashboards and visual outputs support data-driven decision-making by presenting complex environmental data in a simple and interactive format. The system not only measures sustainability performance but also identifies high-risk areas and provides actionable recommendations for carbon reduction, water conservation, and climate resilience planning. This makes the framework practical for policymakers, industries, researchers, and environmental planners.

Overall, the project contributes toward building an integrated sustainability assessment model that aligns with global environmental goals and promotes responsible resource management. By bridging environmental indicators with intelligent analytics and decision support, the system serves as a step toward achieving long-term ecological balance, improved environmental quality, and enhanced human well-being.

5.2 FUTURE ENHANCEMENT

The project can be significantly enhanced by integrating real-time data acquisition systems using IoT sensors, satellite monitoring, and smart environmental devices to enable continuous tracking of water usage, carbon emissions, and climate parameters instead of depending only on historical datasets. This would improve the accuracy, transparency, and responsiveness of sustainability assessments. Additionally, incorporating advanced AI and deep learning models for predictive analytics would allow the system to forecast long-term carbon and water footprint trends, detect anomalies, predict climate-related risks, and recommend optimized sustainability strategies. Such advancements would transform the framework from a descriptive analytical tool into a proactive and intelligent decision-support system.

Furthermore, the framework can be expanded to develop a Global Composite Sustainability Index that integrates water footprint, carbon footprint, climate vulnerability, biodiversity indicators, and human well-being metrics into a unified standardized score for global benchmarking and SDG monitoring. Integration with smart city infrastructure could enable urban sustainability dashboards for tracking traffic emissions, green buildings, renewable energy usage, and waste management

APPENDICES

SAMPLE CODE

```
import sys

import json

import warnings

import os

from pathlib import Path

from typing import Dict, Any, List, Optional, Union, Tuple

from datetime import datetime

import subprocess


warnings.filterwarnings('ignore')


class MLEnsembleSystem:

    """

    Comprehensive ML Ensemble System that integrates all available ML models

    for enhanced accuracy and multi-dimensional analysis

    """

    def __init__(self):

        self.base_path = Path(__file__).parent

        self.models = {

            'basic_lookup': 'predict.py',

            'weather_predictor': 'weather_predictor.py',

            'enhanced_weather_full': 'enhanced_weather_predictor.py',
```

```

    'enhanced_weather_simplified': 'enhanced_weather_predictor_simplified.py',
    'advanced_pipeline': 'advanced_ml_pipeline.py'
}

```

```

def get_basic_footprint(self, product: str) -> Dict[str, Any]:
    """Get basic footprint data using simple lookup model"""
    try:
        result = subprocess.run([
            sys.executable,
            str(self.base_path / self.models['basic_lookup']),
            product
        ], capture_output=True, text=True, cwd=self.base_path.parent)

        if result.returncode == 0:
            data = json.loads(result.stdout)

            # Add product image if not present
            if 'image_url' not in data:
                data['image_url'] = self._get_product_image_url(product)

            return data
        else:
            return {"error": "Basic lookup failed", "details": result.stderr}
    except Exception as e:
        return {"error": "Basic lookup error", "details": str(e)}

```



```
def get_weather_analysis(self, crop: str, city: str = "London", enhanced: bool = True) -> Dict[str, Any]:
```

```
    """Get weather-based analysis using enhanced weather predictors"""
```

```
    try:
```

```
        if enhanced:
```

```
            # Use the simplified enhanced predictor (more stable)
```

```
            result = subprocess.run([
```

```
                sys.executable,
```

```
                str(self.base_path / self.models['enhanced_weather_simplified']),
```

```
                'complete',
```

```
                crop
```

```
            ], capture_output=True, text=True, cwd=self.base_path.parent)
```

```
        else:
```

```
            # Use standard weather predictor
```

```
            result = subprocess.run([
```

```
                sys.executable,
```

```
                str(self.base_path / self.models['weather_predictor']),
```

```
                crop
```

```
            ], capture_output=True, text=True, cwd=self.base_path.parent)
```

```
        if result.returncode == 0:
```

```
            return json.loads(result.stdout)
```

```
        else:
```

```
            return {"error": "Weather analysis failed", "details": result.stderr}
```

```
    except Exception as e:
```

```

        return {"error": "Weather analysis error", "details": str(e)}

def get_advanced_pipeline_analysis(self, operation: str = "pipeline") -> Dict[str, Any]:
    """Get comprehensive analysis using the 5-stage ML pipeline"""
    try:
        result = subprocess.run([
            sys.executable,
            str(self.base_path / self.models['advanced_pipeline']),
            operation
        ], capture_output=True, text=True, cwd=self.base_path.parent)

        if result.returncode == 0:
            return json.loads(result.stdout)
        else:
            return {"error": "Advanced pipeline failed", "details": result.stderr}
    except Exception as e:
        return {"error": "Advanced pipeline error", "details": str(e)}

def ensemble_predict(self, product: str, crop: str = None, city: str = "London") -> Dict[str, Any]:
    """
    Comprehensive ensemble prediction using all available models

    Returns consolidated results with confidence scores
    """
    results = {

```

```

"timestamp": datetime.now().isoformat(),

"product": product,

"crop": crop,

"city": city,

"analysis_type": "Ensemble ML Prediction",

"image_url": self._get_product_image_url(product),

"models_used": [],

"results": {},

"ensemble_summary": {},

"confidence_metrics": {},

"recommendations": []

}

```

1. Basic Footprint Lookup

```

print("Running basic footprint lookup...")

basic_result = self.get_basic_footprint(product)

if "error" not in basic_result:

    results["models_used"].append("Basic Lookup")

    results["results"]["basic_lookup"] = basic_result

```

2. Weather Analysis (if crop is provided)

```

if crop:

    print("Running enhanced weather analysis...")

    weather_result = self.get_weather_analysis(crop, city, enhanced=True)

    if "error" not in weather_result:

```

```

results["models_used"].append("Enhanced Weather Predictor")

results["results"]["weather_analysis"] = weather_result

```

3. Advanced ML Pipeline

```

print("Running advanced ML pipeline...")

pipeline_result = self.get_advanced_pipeline_analysis("pipeline")

if "error" not in pipeline_result:

    results["models_used"].append("5-Stage ML Pipeline")

    results["results"]["advanced_pipeline"] = pipeline_result

```

4. Ensemble Analysis and Consolidation

```

results["ensemble_summary"] = self._create_ensemble_summary(results["results"])

results["confidence_metrics"] = self._calculate_confidence_metrics(results["results"])

results["recommendations"] =
self._generate_ensemble_recommendations(results["results"])

return results

```

```

def _create_ensemble_summary(self, model_results: Dict[str, Any]) -> Dict[str, Any]:

```

```

    """Create consolidated summary from all model results"""

```

```

    summary = {

        "carbon_footprint": {"values": [], "sources": [], "final_estimate": 0},

        "water_footprint": {"values": [], "sources": [], "final_estimate": 0},

        "sustainability_score": 0,

        "accuracy_indicators": {}
    }

```

```

}

# Extract carbon footprint estimates

if "basic_lookup" in model_results:

    carbon = model_results["basic_lookup"].get("carbon_footprint", 0)

    if carbon:

        summary["carbon_footprint"]["values"].append(carbon)

        summary["carbon_footprint"]["sources"].append("Basic Lookup")


    if "weather_analysis" in model_results and "summary" in
model_results["weather_analysis"]:

        carbon =
model_results["weather_analysis"]["summary"].get("predicted_carbon_footprint", 0)

        if carbon:

            summary["carbon_footprint"]["values"].append(carbon)

            summary["carbon_footprint"]["sources"].append("Weather Analysis")


# Extract water footprint estimates

if "basic_lookup" in model_results:

    water = model_results["basic_lookup"].get("water_footprint", 0)

    if water:

        summary["water_footprint"]["values"].append(water)

        summary["water_footprint"]["sources"].append("Basic Lookup")


    if "weather_analysis" in model_results and "summary" in
model_results["weather_analysis"]:

```

```

    water =
model_results["weather_analysis"]["summary"].get("predicted_water_footprint", 0)

    if water:

        summary["water_footprint"]["values"].append(water)

        summary["water_footprint"]["sources"].append("Weather Analysis")


# Calculate ensemble estimates (weighted average)

if summary["carbon_footprint"]["values"]:

    # Give more weight to weather analysis if available

    weights = [2 if "Weather" in source else 1 for source in
summary["carbon_footprint"]["sources"]]

    weighted_sum = sum(v * w for v, w in zip(summary["carbon_footprint"]["values"],
weights))

    total_weight = sum(weights)

    summary["carbon_footprint"]["final_estimate"] = weighted_sum / total_weight


if summary["water_footprint"]["values"]:

    weights = [2 if "Weather" in source else 1 for source in
summary["water_footprint"]["sources"]]

    weighted_sum = sum(v * w for v, w in zip(summary["water_footprint"]["values"],
weights))

    total_weight = sum(weights)

    summary["water_footprint"]["final_estimate"] = weighted_sum / total_weight


# Calculate sustainability score based on available data

if "weather_analysis" in model_results and "summary" in
model_results["weather_analysis"]:

```

```
confidence = model_results["weather_analysis"]["summary"].get("confidence_score",
0.5)
```

```
yield_impact = model_results["weather_analysis"]["summary"].get("yield_impact",
0)
```

```
water_savings =
model_results["weather_analysis"]["summary"].get("potential_water_savings", 0)
```

```
# Sustainability score based on yield impact and water savings
```

```
sustainability = max(0, min(1, (100 + yield_impact) / 100 * confidence +
water_savings / 100))
```

```
summary["sustainability_score"] = sustainability
```

```
return summary
```

```
def _calculate_confidence_metrics(self, model_results: Dict[str, Any]) -> Dict[str, Any]:
```

```
    """Calculate confidence metrics for the ensemble prediction"""
```

```
    metrics = {
```

```
        "overall_confidence": 0.5, # Default
```

```
        "model_agreement": 0,
```

```
        "data_quality": 0,
```

```
        "prediction_reliability": 0
```

```
    }
```

```
# Model agreement - measure consistency between predictions
```

```
carbon_values = []
```

```
water_values = []
```

```

if "basic_lookup" in model_results:

    carbon_values.append(model_results["basic_lookup"].get("carbon_footprint", 0))

    water_values.append(model_results["basic_lookup"].get("water_footprint", 0))


    if "weather_analysis" in model_results and "summary" in
model_results["weather_analysis"]:

        carbon_values.append(model_results["weather_analysis"]["summary"].get("predicted_carbon
_footprint", 0))

        water_values.append(model_results["weather_analysis"]["summary"].get("predicted_water_f
ootprint", 0))


# Calculate coefficient of variation as agreement metric

if len(carbon_values) > 1:

    carbon_mean = sum(carbon_values) / len(carbon_values)

    carbon_std = (sum((x - carbon_mean) ** 2 for x in carbon_values) /
len(carbon_values)) ** 0.5

    carbon_agreement = 1 - (carbon_std / carbon_mean if carbon_mean > 0 else 1)

    metrics["model_agreement"] = max(0, carbon_agreement)


# Data quality based on weather analysis completeness

if "weather_analysis" in model_results:

    weather_data = model_results["weather_analysis"].get("weather_data", {})

    data_completeness = len(weather_data) / 10 # Assume 10 ideal weather parameters

    metrics["data_quality"] = min(1, data_completeness)


# Advanced pipeline reliability

```



```

if "advanced_pipeline" in model_results:

    pipeline_success = len([stage for stage in
model_results["advanced_pipeline"].get("stages", {}).values()

        if stage.get("status") == "success"])

    metrics["prediction_reliability"] = pipeline_success / 5 # 5 stages total


# Overall confidence as weighted average

weights = [0.3, 0.3, 0.4] # agreement, data_quality, reliability

metrics["overall_confidence"] = sum(m * w for m, w in zip(

    [metrics["model_agreement"], metrics["data_quality"],
metrics["prediction_reliability"]],

    weights

))


return metrics


def _generate_ensemble_recommendations(self, model_results: Dict[str, Any]) -> List[str]:

    """Generate actionable recommendations based on ensemble results"""

    recommendations = []


# From weather analysis

    if "weather_analysis" in model_results and "summary" in
model_results["weather_analysis"]:

        weather_recs =
model_results["weather_analysis"]["summary"].get("recommendations", [])

        recommendations.extend(weather_recs[:3]) # Top 3 weather-based recommendations

```

```

# From advanced pipeline

if "advanced_pipeline" in model_results and "stages" in
model_results["advanced_pipeline"]:

    optimization = model_results["advanced_pipeline"]["stages"].get("optimization", {})

    if "optimization" in optimization:

        pipeline_recs = optimization["optimization"].get("recommendations", [])

        recommendations.extend(pipeline_recs[:2]) # Top 2 pipeline recommendations


# General ensemble recommendations

recommendations.append("Consider using ensemble ML predictions for improved
accuracy")

recommendations.append("Monitor confidence metrics to validate prediction
reliability")


return recommendations[:6] # Limit to 6 recommendations


def _get_product_image_url(self, product: str) -> str:

    """Generate product image URL based on product name"""

    # Product image mapping

    product_images = {

        'apple': 'https://images.unsplash.com/photo-1560806887-1e4cd0b6cbd6?w=400&h=300&fit=crop',

        'banana': 'https://images.unsplash.com/photo-1571771894821-ce9b6c11b08e?w=400&h=300&fit=crop',

        'orange': 'https://images.unsplash.com/photo-1547036967-23d11aacaee0?w=400&h=300&fit=crop',

        'tomato': 'https://images.unsplash.com/photo-1546470427-e75e89aaab91?w=400&h=300&fit=crop',
    }

```

'potato': '<https://images.unsplash.com/photo-1518977676601-b53f82aba655?w=400&h=300&fit=crop>',

'carrot': '<https://images.unsplash.com/photo-1445282768818-728615cc910a?w=400&h=300&fit=crop>',

'rice': '<https://images.unsplash.com/photo-1586201375761-83865001e31c?w=400&h=300&fit=crop>',

'wheat': '<https://images.unsplash.com/photo-1574323347407-f5e1ad6d020b?w=400&h=300&fit=crop>',

'corn': '<https://images.unsplash.com/photo-1551754655-cd27e38d2076?w=400&h=300&fit=crop>',

'mango': '<https://images.unsplash.com/photo-1605711285791-0219e80e43a3?w=400&h=300&fit=crop>',

'spinach': '<https://images.unsplash.com/photo-1576045057995-568f588f82fb?w=400&h=300&fit=crop>',

'onion': '<https://images.unsplash.com/photo-1518977822534-7049a61ee0c2?w=400&h=300&fit=crop>',

'garlic': '<https://images.unsplash.com/photo-1471195696269-2d6bf6b5b3b5?w=400&h=300&fit=crop>',

'lettuce': '<https://images.unsplash.com/photo-1556909114-f6e7ad7d3136?w=400&h=300&fit=crop>',

'cucumber': '<https://images.unsplash.com/photo-1449300079323-02e209d9d3a6?w=400&h=300&fit=crop>',

'pepper': '<https://images.unsplash.com/photo-1525607551862-4d197d17c04a?w=400&h=300&fit=crop>',

'broccoli': '<https://images.unsplash.com/photo-1459411621453-7b03977f4bfc?w=400&h=300&fit=crop>',

'cabbage': '<https://images.unsplash.com/photo-1594282486552-05b4d80fbb9f?w=400&h=300&fit=crop>',

'strawberry': '<https://images.unsplash.com/photo-1464965911861-746a04b4bca6?w=400&h=300&fit=crop>',

'grapes': '<https://images.unsplash.com/photo-1537640538966-79f369143f8f?w=400&h=300&fit=crop>'

```

    }

    # Clean product name for lookup

    clean_product = product.lower().strip()

    # Return specific image if available, otherwise return generic food image

    return product_images.get(clean_product, 'https://images.unsplash.com/photo-1498837167922-ddd27525d352?w=400&h=300&fit=crop')

def main():

    """Main function for command-line usage"""

    if len(sys.argv) < 2:

        print(json.dumps({

            "error": "Usage: python ml_ensemble_system.py <product> [crop] [city]",

            "example": "python ml_ensemble_system.py apple wheat London"

        })))

    return

    product = sys.argv[1]

    crop = sys.argv[2] if len(sys.argv) > 2 else None

    city = sys.argv[3] if len(sys.argv) > 3 else "London"

    ensemble = MLEnsembleSystem()

    result = ensemble.ensemble_predict(product, crop, city)

    print(json.dumps(result, indent=2))

```

```

if __name__ == "__main__":
    main()

const express = require('express');
const bodyParser = require('body-parser');
const cors = require('cors');
const footprintRouter = require('./routes/footprint');
const authRouter = require('./routes/auth');
const path = require('path');
const fs = require('fs');

// Load environment configuration
function loadEnvConfig() {
    const envFile = path.join(__dirname, '.env');
    if (fs.existsSync(envFile)) {
        const envContent = fs.readFileSync(envFile, 'utf8');
        envContent.split('\n').forEach(line => {
            line = line.trim();
            if (line && !line.startsWith('#') && line.includes('=')) {
                const [key, value] = line.split('=', 2);
                if (!process.env[key.trim()]) {
                    process.env[key.trim()] = value.trim();
                }
            }
        })
    }
}

```

```

    });

}

}

// Load environment configuration

loadEnvConfig();

const app = express();

const PORT = process.env.PORT || 5000;

// Middleware

app.use(cors({
  origin: ['http://localhost:3000', 'http://127.0.0.1:3000'],
  credentials: true
})); // apply CORS before routes

app.use(bodyParser.json());

app.use(bodyParser.urlencoded({ extended: true }));

// Routes

app.get('/', (req, res) => {
  res.json({
    message: 'Footprint Tracker Backend is running!',
    version: '1.0.0',
    endpoints: {
      footprint: '/api/footprint/:product',

```

```

    weather: '/api/weather/:product',

    enhancedAnalysis: '/api/enhanced-analysis/:crop',

    auth: '/api/auth',

    register: '/api/auth/register',

    login: '/api/auth/login'

  },

  status: 'Production Ready'

});

});

// Use auth router for authentication

app.use('/api/auth', authRouter);

// Use footprint router for API

app.use('/api', footprintRouter);

// Health check endpoint

app.get('/health', (req, res) => {

  res.json({

    status: 'healthy',

    timestamp: new Date().toISOString(),

    uptime: process.uptime()

  });

});

});

```

```

// Start server

app.listen(PORT, () => {

  console.log(`🚀 Footprint Tracker Backend running on http://localhost:${PORT}`);

  console.log(`📊 Environment: ${process.env.NODE_ENV || 'development'}`);

  console.log(`    Weather API: ${process.env.OPENWEATHER_API_KEY ? 'Configured' : 'Using default key'}`);

});

const jwt = require('jsonwebtoken');

const User = require('../models/user');

const JWT_SECRET = process.env.JWT_SECRET || 'your-super-secure-jwt-secret-key-change-this-in-production';

const auth = async (req, res, next) => {

  try {

    // Get token from header

    const authHeader = req.header('Authorization');

    if (!authHeader || !authHeader.startsWith('Bearer ')) {

      return res.status(401).json({

        success: false,

        message: 'Access denied. No token provided.'

      });

    }

  }

```



```

// Extract token

const token = authHeader.substring(7); // Remove "Bearer " prefix


if (!token) {

  return res.status(401).json({

    success: false,

    message: 'Access denied. No token provided.'

  });

}


// Verify token

const decoded = jwt.verify(token, JWT_SECRET);


// Check if user still exists

const user = await User.findById(decoded.userId);


if (!user) {

  return res.status(401).json({

    success: false,

    message: 'Token is no longer valid. User not found.'

  });

}


// Check if account is locked

```

```
if (User.isLocked(user)) {  
  return res.status(423).json({  
    success: false,  
    message: 'Account is temporarily locked.'  
  });  
}
```

```
// Add user info to request  
req.userId = decoded.userId;  
req.user = user;
```

```
next();
```

```
} catch (error) {  
  if (error.name === 'JsonWebTokenError') {  
    return res.status(401).json({  
      success: false,  
      message: 'Invalid token.'  
    });  
  }  
}
```

```
if (error.name === 'TokenExpiredError') {  
  return res.status(401).json({  
    success: false,  
    message: 'Token has expired.'  
  });  
}
```

```
    });  
  }  
  
  console.error('Auth middleware error:', error);  
  res.status(500).json({  
    success: false,  
    message: 'Token verification failed.'  
  });  
}  
};  
  
module.exports = auth;
```

REFERENCES

1. Alotaibi, E., & Nassif, N. (2024): Artificial Intelligence in Environmental Monitoring: In-Depth Analysis Discover Artificial Intelligence It Offers a comprehensive bibliometric review of AI/ML applications in environmental monitoring across air, water, climate, and biodiversity
2. Ding, Z., Wang, J., Song, Y., et al. (2024). Tracking the carbon footprint of global generative artificial intelligence. While this paper focuses on the carbon footprint of AI.
3. Hoekstra, A. Y., Chapagain, A. K., Aldaya, M. M., & Mekonnen, M. M. (2011): The Water Footprint Assessment Manual: Setting the Global Standard. The green, blue and grey water footprint of crops and derived crop products.
4. Ingold, T. (2010). Footprints through the weather-world: Walking, breathing, knowing.
5. Jones, C., & Kammen, D. M. (2018). Spatial distribution of U.S. consumer energy and carbon footprints.
6. Khan et al. (2020): "A machine learning approach for predicting the agricultural water footprint based on climate data."
7. Kumar, P., "Machine Learning Applications in Environmental Sustainability," IEEE Access, 2021.
8. Li, Y., Chen, Y., Yu, X., & Wu, W. (2023). Machine learning modeling of water footprint in crop production distinguishing water supply and irrigation method scenarios. This paper provides a concrete example of how ML is used to model water footprints in a specific sector (agriculture).
9. Mokhtar, A. et al., Estimation of the rice water footprint based on machine learning (2021).
10. Otto, F. E. L. (2017). The science of attributing extreme weather events and its potential contribution to assessing loss and damage associated with climate change impacts,. Attributing extreme weather to climate change: A new frontier in climate science.
11. Peters, G. P. (2010). Carbon footprints and embodied carbon at multiple scales. Current Opinion in Environmental Sustainability.
12. Pfister, S., & Hellweg, S. (2009): The water footprint of electricity from hydropower plants.
13. Ridoutt, B. G., & Pfister, S. (2010). A revised approach to water footprinting to make transparent the impacts of consumption and production on global freshwater scarcity A new water footprint calculation method integrating consumptive and degradative water use into a single stand-alone indicator.